

Neural Networks Basics - Study Notes

Neurons and layers

An artificial neuron computes a weighted sum of its inputs, adds a bias and passes the result through an activation function. Neurons are organized in layers: an input layer, one or more hidden layers and an output layer. A network with many hidden layers is called a deep neural network.

Activation functions

ReLU outputs $\max(0, x)$: it returns zero for negative inputs and the input itself for positive inputs. It is cheap to compute, helps against vanishing gradients and is the default choice for hidden layers. Sigmoid squashes values into the range 0 to 1 and is used for binary outputs. Tanh squashes values into the range -1 to 1. Softmax converts a vector of scores into probabilities that sum to 1 and is used for multi-class outputs.

How networks are trained

Training repeats three steps: a forward pass computes the predictions, the loss function measures the error, and backpropagation computes the gradients of the loss with respect to every weight using the chain rule. Gradient descent then updates the weights. The learning rate controls the step size: too high and training diverges, too low and training is very slow. Mini-batch stochastic gradient descent and the Adam optimizer are widely used.

Epochs and batches

One epoch is one full pass over the training data. The batch size is the number of examples used for a single weight update. Smaller batches give noisier but more frequent updates.

Vanishing gradients

In deep networks that use sigmoid or tanh activations the gradients can become extremely small in the early layers, so those layers barely learn. ReLU activations, residual connections and batch normalization all help to reduce this problem.

Common architectures

Convolutional neural networks (CNNs) are designed for images: convolutional filters detect local patterns such as edges and textures. Recurrent networks such as LSTMs process sequences step by step. Transformers use self-attention to relate all positions in a sequence to each other and are the foundation of modern large language models.